

On the Estimation of the Topp-Leone Distribution

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Revised, October 2026

ABSTRACT

The Topp-Leone distribution is attractive for reliability studies as it has a J-shaped density and a bathtub-shaped hazard function. It also provides the basis for constructing other probability distributions with desirable features. We compare some important properties of various estimators of this distribution's shape parameter when the support is the unit interval. These include the method of moments and maximum likelihood estimators; their bootstrap and jackknife bias-adjusted counterparts; and an analytically bias-adjusted maximum likelihood estimator. The last of these estimators is very simple to apply, and it dominates the other estimators in terms of both relative bias and mean squared error in finite samples. The theoretical results are illustrated with three applications involving actual data.

Keywords: Bias reduction; maximum likelihood; method of moments; unbiased estimation; mean squared error; bathtub hazard; J-shaped distribution; finite support

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1. Introduction

The range of distributions whose support is finite (and especially the unit interval) has increased dramatically in recent years. There have been numerous applications across many fields and topics, such as electricity output capacity (Mazumdar and Gaver, 1984 and Zghoul, 2020); income distribution (Bayoud, 2016a); poverty (Rodrigues *et al.*, 2020); anxiety performance (Okorie *et al.*, 2017); literacy rates and other well-being indicators (Guerra *et al.*, 2021); medical issues (Shekhawat and Sharma, 2021 and Suleiman *et al.*, 2024); tensile strength of fibers (Quesenberry and Hales, 1980); engineering and environmental data (Bakouch *et al.*, 2023). In fact, Afuecheta *et al.* (2025) summarize and discuss the details of more than 120 distributions whose support is the $[0,1]$ interval.

In this paper we study the sampling properties of the maximum likelihood (ML) and method of moments (MOM) estimators of the shape parameter of an important distribution that has finite support and was first introduced by Topp and Leone (1955). Nadarajah and Kotz (2003) “re-discovered” the Topp-Leone (T-L) distribution, and it has attracted recent attention – *e.g.*, Ghitany *et al.* (2005), Van Dorp and Kotz (2006), Zhou *et al.* (2006), Kotz and Seier (2007), Nadarajah (2009), Al-Zahrani (2012), Genç (2012, 2013), and Benkhelifa (2017). As well as having finite support, the T-L distribution has a uni-modal density with a shape parameter, v . Moreover, the density function is “J-shaped”, the hazard function is “bathtub-shaped”, and the mean residual life function is “upside-down-bathtub shaped” for all values of $v \in (0, 1)$. This unusual characteristic of the hazard function is especially important in reliability applications in a wide range of fields, as is discussed by Reed (2011), for example.

For a given support, the shape parameter of the T-L distribution is easily estimated by ML or MOM. We show that both of these estimators are positively biased in finite samples. The ML estimator can be expressed in closed form, so it is especially simple to compute. It exhibits greater (relative) bias than the method of moments estimator, but smaller relative mean squared error. We derive the analytic bias of the ML estimator to $O(n^{-1})$, and prove that the corresponding bias-corrected estimators based on the different approaches of Cox and Snell (1968) and Firth (1993) are identical for this problem. A simulation experiment illustrates that this bias-corrected ML estimator dominates the MOM estimator in terms of both (relative) bias and mean squared error. In addition, it is superior to both the bootstrap bias-corrected and jackknife bias-corrected versions of the MOM and ML estimators, and is very simple to construct. Finally, three applications are presented to illustrate our new results with real data.

2. The Topp-Leone distribution

If X follows the T-L distribution with support $(0, b)$, and shape parameter v , its distribution function is:

$$F(x) = (x/b)^v (2 - x/b)^v \quad ; \quad 0 < x < b < \infty ; \quad v > 0 \quad (1)$$

and its density function is

$$f(x) = (2v/b) (x/b)^{(v-1)} (1 - x/b) (2 - x/b)^{(v-1)} . \quad (2)$$

Nadarajah and Kotz (2003) establish the moments of this distribution, and those of its order statistics are discussed by Genç (2012) and Mirmostafaei (2014). Importantly, and in contrast to some other finite-support distributions (such as the beta distribution), the hazard function for the T-L distribution has a simple closed form.

Without loss of generality, in this study we follow the majority of the literature associated with this distribution and assign $b = 1$. This is an especially interesting case. Not only is the unit-interval support relevant when modelling data measuring proportions, but it is also associated with the generation of important new families of distributions. For the T-L distribution this is discussed by Al-Shomrani *et al.* (2016), who extend earlier studies in which the beta and Kumaraswamy (1980) distribution functions are used for this purpose. Subsequent extensions involving the T-L distribution are provided by various authors, including Sangsanit and Bodhisuwan (2016), Rezaei *et al.* (2017), Abbas *et al.* (2017), Sharma (2018), Oguntunde *et al.* (2019), Sudsila *et al.* (2022), and Ehizojie (2026). Accordingly, (1) and (2) simplify to:

$$F(x) = x^v (2 - x)^v \quad ; \quad 0 < x < 1 ; \quad v > 0 \quad (3)$$

and

$$f(x) = 2vx^{v-1}(1 - x)(2 - x)^{v-1} , \quad (4)$$

and the hazard function is:

$$\lambda(x) = 2vy(1 - y^2)^{v-1} / [1 - (1 - y^2)^v] , \quad (5)$$

where $y = (1 - x)$.

The J-shape of the T-L density, and the bathtub shape of the hazard function (for all $v \in (0,1)$), are of particular interest, and are illustrated by Nadarajah and Kotz (2003, pp. 312-313). In contrast, if $v > 1$, the densities in (2) and (4) are still proper, all of its integer-order moments exist, but the density is not J-shaped, and the corresponding hazard function no longer has a bathtub shape. See Kotz and Seier (2007, pp. 6-10) for a discussion of the characteristics of the distribution when $1 < v < 2$, and when $v > 2$, and why the T-L distribution is essentially a generalization of the left-triangular distribution.

Under independent sampling, with sample size n , the log-likelihood function based on (4) is:

$$l = n \log(2) + n \log(v) + \sum_{i=1}^n [(v-1) \log(x_i) + \log(1-x_i) + (v-1) \log(2-x_i)] . \quad (6)$$

Noting that

$$\partial l / \partial v = (n/v) + \sum_{i=1}^n [\log(x_i) + \log(2-x_i)] , \quad (7)$$

and

$$(\partial^2 l) / (\partial v^2) = -n/v^2 < 0, \quad (8)$$

the maximum likelihood estimator (MLE) for v can be expressed in closed-form as

$$\hat{v} = -n / \sum_{i=1}^n \log[x_i(2-x_i)], \quad (9)$$

and so $\hat{v} > 0$, as $0 < x_i < 1$ for all i . The T-L density satisfies the usual regularity conditions, so the MLE of v is weakly consistent and best asymptotically normal. However, its finite-sample properties are not readily deduced, given that the estimator is a highly non-linear function of the data.

Bayoud (2016a, p.74) shows that this MLE coincides with the mean of the posterior density (which is gamma in form) in a Bayesian analysis of this problem with a non-informative prior. Accordingly, the ML estimator is also the Bayes estimator of v when the loss function is quadratic. Feroze and Aslam (2013), Sindhu *et al.* (2013) and Sultan and Ahmad (2016) also consider Bayesian inference in the context of the Topp-Leone distribution, under various sampling schemes, priors and loss functions. ML estimation of the shape parameter under censored sampling is discussed, for example, by Feroze and Aslam (2013), Bayoud (2015, 2016b) and Long (2022); and Zghoul (2020) considers various plug-in estimators for this parameter. The relative efficiency of ML-based estimation of the T-L density and distribution functions is established by Benkhelifa (2017).

The MOM estimator of v is obtained by solving the equation, $E(X) = \bar{x} = n^{-1} \sum_{i=1}^n x_i$, for v . From Nadarajah and Kotz (2003; p.315), $E(X) = 1 - 4^v (\Gamma(1+v))^2 / \Gamma(2+2v)$, so the MOM estimator ($\tilde{v}$) is obtained as the solution to the nonlinear equation,

$$\Gamma(2+2v)(\bar{X} - 1) + 4^v (\Gamma(1+v))^2 = 0 . \quad (10)$$

Although the MOM estimator is also weakly consistent for v , its finite-sample properties have not been explored previously. While the function of v in (10) is non-trivial, straightforward numerical evaluations

show that it has a unique (positive) root for all $0 < \bar{x} < 1$, and hence for our situation where $0 < x_i < 1$; $i = 1, 2, \dots, n$. (Also, see Zghoul, 2020, p.308.)

By construction, $\hat{v} > 0$ and $\tilde{v} > 0$, but it is possible for these estimators to yield estimates that exceed unity, especially for very small n , even if $v \in (0,1)$. This, in itself, is not problematic.

3. Bias-adjusted estimators

One would anticipate the ML and MOM estimators of v may each be biased in finite samples. If so, this could be of concern when the T-L distribution is applied to reliability problems. Here, we investigate this issue, and consider bias-correction strategies for the MLE and MOM estimator.

Cox and Snell (1968) provided a framework for estimating the bias, to $O(n^{-1})$ for MLEs of the parameters of “regular” densities. Then, subtracting the estimated bias from the original MLE produces a bias-corrected estimator that is unbiased to $O(n^{-2})$. This type of “corrective” bias adjustment has been applied successfully in many contexts. Recent examples include Cordeiro and Klein (1994), Lemonte *et al.* (2007), Lemonte (2011), Giles *et al.* (2013), Schwartz *et al.* (2013), Xiao and Giles (2014), Schwartz and Giles (2016), and Giles *et al.* (2016).

In general terms, suppose that the parameter vector, θ , is of dimension p . Defining

$$k_{ij} = E(\partial^2 l / \partial \theta_i \partial \theta_j) \quad ; \quad i, j = 1, 2, \dots, p \quad (11)$$

$$k_{ijl} = E(\partial^3 l / \partial \theta_i \partial \theta_j \partial \theta_l) \quad ; \quad i, j, l = 1, 2, \dots, p \quad (12)$$

$$k_{ij,l} = E[(\partial^2 l / \partial \theta_i \partial \theta_j)(\partial l / \partial \theta_l)] \quad ; \quad i, j, l = 1, 2, \dots, p \quad (13)$$

Cox and Snell (1968) showed that the bias of the s^{th} element of the ML estimator of θ ($\hat{\theta}$) is:

$$\text{Bias}(\hat{\theta}_s) = \sum_{i=1}^p \sum_{j=1}^p \sum_{l=1}^p k^{si} k^{jl} [0.5k_{ijl} + k_{ij,l}] + O(n^{-2}); \quad s = 1, 2, \dots, p, \quad (14)$$

where k^{ij} is the $(i,j)^{\text{th}}$ element of the inverse of the information matrix, $K = \{-k_{ij}\}$.

Cordeiro and Klein (1994) note that this bias expression can be re-written as:

$$\text{Bias}(\hat{\theta}_s) = \sum_{i=1}^p k^{si} \sum_{j=1}^p \sum_{l=1}^p [k_{ij}^{(l)} - 0.5k_{ijl}] k^{jl} + O(n^{-2}); \quad s = 1, 2, \dots, p \quad (15)$$

where

$$k_{ij}^{(l)} = \partial k_{ij} / \partial \theta_l \quad ; \quad i, j, l = 1, 2, \dots, p. \quad (16)$$

The computational advantage of (15) over (14) is that it avoids computing expectations of products (see (13)). Then, defining $a_{ij}^{(l)} = k_{ij}^{(l)} - (k_{ijl} / 2)$, for $i, j, l = 1, 2, \dots, p$; and constructing the matrices:

$$A^{(l)} = \{a_{ij}^{(l)}\}; \quad i, j, l = 1, 2, \dots, p \quad (17)$$

$$A = [A^{(1)} | A^{(2)} | \dots | A^{(p)}], \quad (18)$$

Cordeiro and Klein (1994) show that the expression for the $O(n^{-1})$ bias of $\hat{\theta}$ can be re-written as:

$$Bias(\hat{\theta}) = K^{-1} A vec(K^{-1}) + O(n^{-2}). \quad (19)$$

Here, $K = \{-k_{ij}\}$, is the (expected) Fisher information matrix. Then, a “bias-corrected” ML estimator for θ can then be constructed as:

$$\hat{\theta}_{CS} = \hat{\theta} - \hat{K}^{-1} \hat{A} vec(\hat{K}^{-1}), \quad (20)$$

where $\hat{K} = (K)|_{\hat{\theta}}$ and $\hat{A} = (A)|_{\hat{\theta}}$.

In our problem, $p = 1$, $K = (n/v^2)$, $(\partial^3 l)/(\partial v^3) = k_{111} = k_{11}^{(1)} = (2n/v^3)$, and $A = a_{11}^{(1)} = (n/v^3)$. So, from (16), $Bias(\hat{v}) = (v/n) + O(n^{-2})$, and the Cox-Snell bias-corrected estimator is

$$\hat{v}_{CS} = \hat{v} - (\hat{v}/n) = (n-1)\hat{v}/n, \quad (21)$$

where $\hat{v}$ is defined in (8).

Firth (1993) suggested an alternative “preventive” approach to bias correction that involves adjusting the score vector before solving the likelihood equation(s) for the MLE. In the present context and notation, Firth’s estimator requires that we solve the equation

$$U^* = (\partial l / \partial v) - A vec(K^{-1}) = 0. \quad (22)$$

Using (4), and the expressions for K and A , we immediately obtain the solution:

$$\begin{aligned} \hat{v}_F &= (1-n) / \sum_{i=1}^n [\log(x_i) + \log(2-x_i)] \\ &= (1-n) / \sum_{i=1}^n [1 - (x_i - 1)^2] = \hat{v}_{CS} \end{aligned} \quad (23)$$

So, for this particular estimation problem, the Cox-Snell and Firth bias-corrected MLEs are identical, and henceforth it will be identified as $\hat{v}_{CSF}$. Moreover, $\hat{v}_{CSF}$ is also a particular Bayes estimator. Bayoud (2016a) shows that under a non-informative prior the posterior for v is a gamma distribution with a shape parameter of n , and a scale parameter of $-1/\ln T$, where $T = \prod_{i=1}^n [x_i(2 - x_i)]$. If $\ln T \geq -1$, the mode of this distribution is $(1 - n)/\ln T = \hat{v}_{CSF}$. This is the Bayes estimator of v under a “zero-one” loss function.

4. Simulation experiment

The (percentage) biases and mean squared errors of the MLE, bias-adjusted MLE, and MOM estimator of v have been investigated in a small simulation experiment. This experiment was conducted using the R software environment (R Core Development Team, 2016). The MOM estimator was obtained by solving equation (8) using the *uniroot* function in base R. The Monte Carlo simulation involved $N = 50,000$ replications, and the T-L random variates were generated *via* the “rtopple” routine in the ‘VGAM’ package in R (Yee, 2020). The quality of the random variates was verified by comparing the first four empirical moments of a sample of 10,000 simulated values with the corresponding population moments, as given by Nadarajah and Kotz (2003, p.315).

The simulation results appear in Table 1. There, if $\hat{v}_{(i)}$ is the MLE of v obtained from the i^{th} replication of the experiment, then the percentage bias and mean squared error are $\%Bias(\hat{v}) = 100[(1/N) \sum_{i=1}^N \hat{v}_{(i)} - v]/v$, and $\%MSE(\hat{v}) = 100[(1/N) \sum_{i=1}^N (\hat{v}_{(i)} - v)^2]/v^2$, respectively. It transpires that the percentage biases and MSEs of the MLE and bias-corrected MLE of v are invariant to the value of v itself. In contrast, the relative bias and relative MSE of the MOM estimator of v depend on the value of that parameter. We see in Table 1 that the MOM estimator exhibits less relative bias, but much greater percentage MSE than does the MLE of v . However, the simple Cox-Snell/Firth bias correction is extremely effective. The positive relative bias of the adjusted MLE is much less than that of the MOM estimator of v (typically by an order of magnitude); and in addition, bias-correcting the MLE also reduces the $\%MSE$ of that estimator slightly.

Two alternative ways to correct for estimator for its first-order bias are *via* the parametric bootstrap (*e.g.*, Efron, 1979), and by applying the (“delete one”) jackknife (*e.g.*, Quenouille, 1956). Efron and Tibshirani (1994) establish the asymptotic equivalence of these two bias corrections, but in small samples they may yield quite different results. In addition, Reeds (1978) proves that arbitrary ML estimators behave appropriately when jackknifed. Our simulation study is extended to include each of these bias-corrective procedures for both the ML and MOM estimators. The bootstrap bias-corrected

MLE estimator of v , based on $N_B = 1,000$ bootstrap draws, is obtained as $\hat{v}_B = 2\hat{v} - N_B^{-1} \sum_{k=1}^{N_B} \hat{v}_k$, where $\hat{v}_k$ is the MLE based on the k^{th} bootstrap sample. The bootstrap bias-corrected MOM estimator, $\tilde{v}_B$ is defined in an analogous manner. Further, let $\hat{v}^{(j)}$ be the MLE of v based on the sample with the j^{th} observation omitted ($j = 1, 2, \dots, n$). Then the “delete one” jackknife MLE is $\hat{v}_{jack} = n^{-1} \sum_{j=1}^n \hat{v}^{(j)}$, and the jackknife-biased-corrected MLE is $\hat{v}_J = n\hat{v} - (n-1) \hat{v}_{jack}$. Its MOM counterpart is $\tilde{v}_J = n\tilde{v} - (n-1) \tilde{v}_{jack}$, where $\tilde{v}_{jack} = n^{-1} \sum_{j=1}^n \tilde{v}^{(j)}$ and $\tilde{v}^{(j)}$ is the MOM of v based on the sample with the j^{th} observation omitted ($j = 1, 2, \dots, n$). This is implemented using the ‘jackknife’ function in the ‘bootstrap’ package in R (Tibshirani and Leisch, 2019).

Again, 50,000 simulation replications were used, and the bootstrap and jackknife results are presented in Table 2. There, we see that although the bootstrap, jackknife, and analytically bias-corrected MLEs have similar %MSE’s, the latter estimator (in Table 1) dominates the other two bias-corrected estimators in terms of relative bias, especially for small n . In addition, the jackknife bias-corrected MLE is slightly superior to its bootstrap-corrected counterpart in terms of relative bias. The performances of the various bias-corrected MOM estimators depend in part on the (unknown) value of the parameter being estimated. The jackknife-corrected MOM performs somewhat better than the bootstrap-corrected MOM in terms of relative bias, but their performances in terms of %MSE are similar. Finally, for moderate sample sizes, and especially for small values of v , both the jackknife and bootstrap bias-corrected MLEs dominate their MOM counterparts. In summary, in terms of both relative bias and MSE, the easy-to-apply Cox-Snell/Firth bias-corrected ML estimator is clearly recommended, even for very small sample sizes.

5. Empirical applications

We have conducted three empirical applications to illustrate the consequences of using the various bias-corrected estimators for the Topp-Leone distribution with actual data. The first application involves data from Mazumdar and Gaver (1984) and analyzed previously by Altun and Hamedani (2018). The 23 sample values measure unit capacity factors, using the so-called “SC16” algorithm. The second application uses data provided in Example 2 of Wang (2000), and analyzed by various other authors. These data relate to the life-spans of 18 electronic devices, and their values were divided by 1,000 in order to lie in the unit interval. The data for the third application measure survival times for leukemia patients (Feigl and Zelen, 1965), divided by 56 weeks. The estimation results are presented in Table 3.

Based on the power results described by Al-Zahrani (2012), the Kolmogorov-Smirnov (K-S) test was chosen for goodness-of-fit, and implemented with the ‘dgof’ package in R (Arnold and Emerson, 2024). The Topp-Leone model is supported by the finite sample p-values for the K-S test (reported in Table 1 of Al-Zahrani, 2012) in all cases where an ML-based estimator is used, although not fully in the second application when MOM estimation is adopted. Moreover, the smallest calculated Kolmogorov-Smirnov statistics values are associated with the Cox-Snell-Firth estimator in two of the three applications.

As might be expected from the sample sizes and the bias results in Tables 1 and 2, all of the bias-corrected ML and MOM estimates in Table 3 are smaller than their “uncorrected” counterparts. These reductions range from 3.2% to 6.3% in value in the case of the MLEs, with the largest adjustment always associated with the Cox-Snell-Firth estimator. In the case of the MOM estimates the reductions in the estimate values range from 1.4% to 6.8%, with similar results for both the bootstrap and jackknife bias-corrected estimates. In the first two applications the MOM estimates are noticeably different from their ML counterparts. The value of the “unadjusted” MOM estimate exceeds that of the “unadjusted” ML estimate by 31% in the case of the SC16 data application. In contrast, the “unadjusted” ML estimate exceeds its MOM counterpart by 38% in application with the electronic device data.

All of the estimates of the shape parameter in the three applications lie in the unit interval, so the fitted density and hazard functions are J-shaped and bathtub-shaped respectively. However, the differences in the shape estimates have important implications for the implied kurtosis measures, as is discussed in detail by Kotz and Seier (2007). The Pearson kurtosis estimates included in Table 3 are computed using the formulae for the second and fourth central moments provided by Nadarajah and Koz (2003, p.316), and the kurtosis equals the value of 3 when the shape parameter equals 0.6283063. In that table we see that bias-correcting always increases the kurtosis of the fitted T-L distribution, by up to 3.8%. In addition, the results associated with the first application demonstrate that choosing between ML and MOM estimation can affect whether the estimated kurtosis is greater or less than the “normal” value of 3.

6. Conclusions

Recently, there has been renewed interest in the Topp-Leone distribution, especially in relation to reliability studies, where its finite support and bathtub-shaped hazard function are very appealing to practitioners, and in the construction of other distributions. For a fixed support, method of moments estimation of the distribution’s shape parameter is straightforward enough, but the maximum likelihood estimator is trivial to compute. Correcting the latter estimator to remove its $O(n^{-1})$ bias analytically is

also very straightforward. This analytically bias-corrected estimator is attractive because it has smaller percentage bias and mean squared error than the method of moments estimator in samples of the size likely to be encountered in practice. It also has a natural Bayesian interpretation, and in terms of both percentage bias and percentage mean squared error it out-performs the method of moments and maximum likelihood estimators when these are bias-corrected using either the parametric bootstrap or the delete-one jackknife.

Acknowledgment: I am grateful to Ryan Godwin, Helen Feng, and Joseph Schwartz for their helpful comments on a much earlier version of this work.

Conflicts of Interest: The author declares no conflicts of interest.

Funding and Support: This research received no external funding.

Data Availability: The data used in the applications can be downloaded from <https://github.com/DaveGiles1949/Data>. The R code for the simulations and applications can be downloaded from <https://github.com/DaveGiles1949/r-code>.

Table 1: % Biases [and %MSEs] of ML, bias-corrected ML and MOM estimators

n	$\%Bias(\hat{v})$ [%MSE($\hat{v}$)]	$\%Bias(\hat{v}_{CSF})$ [%MSE($\hat{v}_{CSF}$)]	$\%Bias(\tilde{v})$ [%MSE($\tilde{v}$)]				
$v :$	> 0	0.10	0.25	0.50	0.75	0.90	
10	11.031 [16.516]	-0.072 [12.392]	6.308 [84.247]	6.449 [41.903]	6.773 [28.133]	7.050 [23.760]	7.190 [22.364]
25	4.049 [5.238]	-0.113 [4.329]	2.110 [29.288]	2.232 [14.330]	2.239 [9.420]	2.493 [7.832]	2.549 [7.318]
50	2.000 [2.206]	-0.040 [2.081]	0.938 [14.303]	1.029 [6.917]	1.126 [4.499]	1.194 [3.718]	1.226 [3.465]
75	1.333 [1.412]	-0.018 [1.358]	0.669 [9.401]	0.716 [4.548]	0.774 [2.950]	0.814 [2.430]	0.833 [2.261]
100	1.001 [1.052]	-0.009 [1.021]	0.538 [7.073]	0.568 [3.416]	0.604 [2.215]	0.631 [1.824]	0.645 [1.697]
250	0.379 [0.409]	-0.022 [0.404]	0.193 [2.817]	0.191 [1.357]	0.203 [0.877]	0.218 [0.721]	0.222 [0.670]
500	0.190 [0.202]	-0.011 [0.201]	0.108 [1.400]	0.093 [0.676]	0.094 [0.438]	0.099 [0.360]	0.102 [0.334]
1000	0.099 [0.101]	-0.001 [0.100]	0.095 [0.698]	0.076 [0.338]	0.068 [0.219]	0.067 [0.180]	0.067 [0.167]

Table 2: %Biases [and %MSEs] of bootstrap and jackknife bias-corrected ML and MOM estimators

	% Bias [% MSE]		% Bias [% MSE]		% Bias [% MSE]		% Bias [% MSE]		% Bias [% MSE]		% Bias [% MSE]	
	$\hat{v}_B$	$\hat{v}_J$	$\tilde{v}_B$	$\tilde{v}_J$	$\tilde{v}_B$	$\tilde{v}_J$	$\tilde{v}_B$	$\tilde{v}_J$	$\tilde{v}_B$	$\tilde{v}_J$	$\tilde{v}_B$	$\tilde{v}_J$
v :	> 0		0.10		0.25		0.5		0.75		0.90	
n												
10	-0.265 [12.743]	-1.305 [13.334]	-1.224 [67.828]	-0.121 [69.533]	-1.267 [32.922]	-0.348 [33.739]	-1.312 [21.550]	-0.509 [22.055]	-1.330 [17.929]	-0.603 [18.323]	-1.332 [16.773]	-0.644 [17.127]
25	0.243 [4.385]	-0.267 [4.329]	-0.258 [27.719]	0.308 [28.096]	-0.146 [13.481]	0.106 [13.548]	-0.075 [8.780]	-0.011 [8.765]	-0.046 [7.247]	-0.062 [7.214]	-0.036 [6.748]	-0.082 [6.710]
50	0.080 [2.120]	-0.081 [2.081]	-0.062 [14.021]	0.321 [13.972]	-0.016 [6.779]	0.137 [6.742]	0.016 [4.403]	0.054 [4.366]	0.024 [3.631]	0.025 [3.595]	0.027 [3.379]	0.014 [3.343]
75	0.016 [1.373]	-0.068 [1.363]	-0.002 [9.228]	0.138 [9.276]	-0.017 [4.459]	0.070 [4.480]	-0.029 [2.886]	0.031 [2.897]	0.031 [2.376]	0.013 [2.381]	-0.031 [2.210]	0.006 [2.213]
100	0.016 [1.031]	-0.041 [1.022]	0.108 [7.022]	0.147 [6.950]	0.053 [3.379]	0.080 [3.358]	0.026 [2.187]	0.040 [2.176]	0.019 [1.799]	0.024 [1.792]	0.016 [1.673]	0.018 [1.666]
250	-0.002 [0.401]	0.005 [0.400]	-0.149 [2.788]	0.101 [2.787]	-0.098 [1.340]	0.066 [1.347]	-0.064 [0.865]	0.043 [0.869]	-0.048 [0.711]	0.033 [0.714]	-0.042 [0.661]	0.029 [0.663]
500	0.015 [0.200]	0.028 [0.200]	0.033 [1.406]	0.062 [1.389]	0.035 [0.673]	0.041 [0.670]	0.032 [0.433]	0.031 [0.433]	0.029 [0.355]	0.027 [0.357]	0.028 [0.330]	0.028 [0.331]
1000	0.017 [0.100]	0.026 [0.101]	-0.023 [0.694]	0.034 [0.693]	0.004 [0.334]	0.028 [0.335]	0.010 [0.215]	0.024 [0.217]	0.011 [0.177]	0.024 [0.181]	0.012 [0.164]	0.023 [0.166]

Table 3: Empirical application results*

	$\hat{\nu}$	$\hat{\nu}_{CSF}$	$\hat{\nu}_B$	$\hat{\nu}_J$	$\tilde{\nu}$	$\tilde{\nu}_B$	$\tilde{\nu}_J$
Application 1: SC16 data ($n = 23$)							
Estimates:	0.594 (0.118)	0.568 (0.113)	0.575 (0.118)	0.574 (0.107)	0.780 (0.309)	0.740 (0.309)	0.741 (0.309)
Kurtosis:	3.106	3.198	3.173	3.177	2.661	2.734	2.733
K-S:	0.217	0.174	0.261	0.261	0.217	0.261	0.261
Application 2: Electronic device data ($n = 18$)							
Estimates:	0.598 (0.112)	0.565 (0.106)	0.575 (0.112)	0.579 (0.109)	0.370 (0.088)	0.360 (0.088)	0.365 (0.087)
Kurtosis:	3.093	3.211	3.174	3.157	4.410	4.516	4.461
K-S:	0.167	0.167	0.278	0.222	0.333**	0.278	0.444***
Application 3: Leukemia data ($n = 15$)							
Estimates:	0.750 (0.152)	0.703 (0.142)	0.729 (0.152)	0.724 (0.134)	0.769 (0.358)	0.731 (0.358)	0.717 (0.358)
Kurtosis:	2.715	2.810	2.755	2.766	2.680	2.752	2.779
K-S:	0.250	0.250	0.312	0.313	0.250	0.188	0.313

* Bootstrapped standard errors, based on 1,000 replications, appear in parentheses; K-S = Kolmogorov-Smirnov test statistic, all but two of which have finite-sample p-values exceeding 25%.

** The associated finite-sample p-value lies in [0.10, 0.20].

*** The associated finite-sample p-value lies in [0.00, 0.01].

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